White Noise Tests

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Introduction

After fitting a time-series model — ARIMA, ETS, regression with ARIMA errors — the residuals should look like white noise: zero mean, constant variance, and no autocorrelation. White-noise tests check these properties formally and are the standard residual-diagnostic battery for time-series modelling. Significant rejection indicates the model has missed structure that should motivate refinement; failure to reject supports model adequacy.

Prerequisites

A working understanding of ARIMA models, residual diagnostics, and the autocorrelation function (ACF) as the visual counterpart to formal autocorrelation tests.

Theory

Three test families dominate practice:

- **Portmanteau tests** (Box-Pierce, Ljung-Box) aggregate squared sample autocorrelations over the first h lags into a single chi-squared statistic. Under the white-noise null, $Q \sim \chi^2_{h-p-q}$ where $p+q$ is the number of fitted ARMA parameters. The Ljung-Box variant has a small-sample correction and is preferred over Box-Pierce.
- **Turning-points test**: counts the number of local maxima and minima in the residual series; under iid, has known mean $2(n-2)/3$ and variance, giving an approximate Normal test for independence.
- **Runs test**: counts runs of consecutive same-sign residuals; tests independence without distributional assumptions.

A failure on Ljung-Box at any lag indicates missed autocorrelation that the model should address; a failure on the runs or turning-points test indicates more general non-randomness.

Assumptions

Null: residuals are iid white noise. The tests have different power against autocorrelation, heteroscedasticity, and non-linearity; no single test detects all departures, so multiple tests are run as a battery.

R Implementation

```r
library(forecast)

set.seed(2026)
fit <- arima(AirPassengers, order = c(0, 1, 1), seasonal = c(0, 1, 1))
resid_fit <- residuals(fit)

# Ljung-Box
Box.test(resid_fit, lag = 24, type = "Ljung-Box", fitdf = 2)
```

```r
# Visual + ARIMA diagnostic
checkresiduals(fit)
```

Output & Results

`Box.test()` returns the test statistic and p-value; `checkresiduals()` aggregates the Ljung-Box test with ACF, histogram, and time-series plots into a single diagnostic display.

Interpretation

A reporting sentence: "Ljung-Box at lag 24 on the residuals from the ARIMA(0,1,1)(0,1,1)

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fit gave $Q = 28.1$ on 22 degrees of freedom ($p = 0.17$); the ACF showed no isolated significant lags, the histogram was approximately Normal, and the residual time-series plot showed no remaining structure. Residuals are consistent with white noise." Always pair the numerical test with the ACF plot.

Practical Tips

- Choose the maximum lag h around $\log n$ to a small multiple thereof; for seasonal models, set h at least one full seasonal period to detect missed seasonal autocorrelation.
- Use `fitdf = p + q` for ARMA(p, q) fitted models; the test ignores the parameter cost otherwise and is anti-conservative.
- A significant Ljung-Box after model fitting indicates missed structure; add AR or MA terms, seasonal terms, or consider a different model family.
- Pair the formal Ljung-Box test with a visual ACF inspection; isolated significant lags can be substantively meaningful even if the omnibus test passes.
- `forecast::checkresiduals()` is the convenient one-call diagnostic battery; use it routinely after every time-series fit.
- For non-Normal residuals, a Jarque-Bera or Shapiro-Wilk test on the residuals complements the ACF-based tests.

R Packages Used

Base R `Box.test()` for Ljung-Box and Box-Pierce; `forecast::checkresiduals()` for the integrated diagnostic; `tseries::runs.test()` for the runs test; `randtests::turning.point.test()` for the turning-points test; `feasts::ljung_box()` for the modern tidyverts implementation.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Time series basics